Etomidate: Drug information - UpToDate

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Contributor Disclosures

For additional information see ”Etomidate: Patient drug information” and ”Etomidate: Pediatric drug information”

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Amidate

Brand Names: Canada

- Tomvi

Pharmacologic Category

- Cortisol Synthesis Inhibitor;
- General Anesthetic

Dosing: Adult

Cushing syndrome

Cushing syndrome (off-label use): Note: Studies have not reported sedation at these etomidate doses; however, patients should be managed in an intensive care unit with sedation scoring every 2 hours initially for the first 24 hours, then every 12 hours. Cortisol levels should be measured every 4 to 6 hours (Ref). Optimal regimen and doses have not been identified; refer to institutional protocols.

IV: Initial: 3 to 5 mg loading dose, followed by 0.02 to 0.05 mg/**kg**/hour continuous IV infusion (usually equates to 1.5 to 4 mg/hour). Titrate to serum cortisol of 18 to 29 mcg/dL (500 to 800 nmol/L) in a physiologically stressed patient **or** 5.5 to 11 mcg/dL (150 to 300 nmol/L) in a nonphysiologically stressed patient. For complete blockade, titrate infusion rate to achieve a cortisol level <5.5 mcg/dL (<150 nmol/L). Hydrocortisone IV is required if complete blockade desired rather than partial blockade (‘block and replace’) (Ref).

General anesthesia

General anesthesia: IV: Initial: 0.3 mg/kg over 30 to 60 seconds for induction of anesthesia; usual dosage range: 0.2 to 0.6 mg/kg.

Procedural sedation

Procedural sedation (off-label use): IV: Initial: 0.1 to 0.2 mg/kg, followed by 0.05 mg/kg every 3 to 5 minutes as needed (Ref).

Rapid sequence intubation outside the operating room

Rapid sequence intubation outside the operating room (induction): Note: Provides no analgesic effects (Ref).

IV: Initial: 0.3 mg/kg; usual dosage range: 0.2 to 0.3 mg/kg (Ref).

Supplementation to subpotent anesthetic agents

Supplementation to subpotent anesthetic agents: IV: Administer smaller increments during short operative procedures to supplement subpotent anesthetic agents, such as nitrous oxide; individualize dosage (usually smaller than the original induction dose).

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; use with caution, risk of toxicity is greater in patients with renal impairment.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1 and 2 obesity (BMI 30 to 39 kg/m^2): IV: Use **actual body weight** for weight-based dose calculations (Ref). Refer to adult dosing for indication-specific doses.

Class 3 obesity (BMI ≥ 40 kg/m^2): IV: Use **adjusted body weight** for weight-based dose calculations (Ref). Refer to adult dosing for indication-specific doses.

Rationale for recommendations: Etomidate has a large V_d (2.5 to 4.5 L/kg) with peak anesthetic effects occurring within minutes and a rapid loss of effect related to redistribution (Ref). In 1 study evaluating patients with a BMI ≥40 kg/m^2, the total dose required to achieve sedation was similar between the observed dose and ideal body weight predicted dose or adjusted body weight predicted dose. The adjusted body weight predicted dose was slightly higher than the observed dose (26.8 vs 21.7 mg) (Ref). Given that the risk of underdosing (eg, awake while paralyzed) outweighs the risk of toxicity, use actual body weight in patients with a BMI 30 to 39 kg/m^2 and consider adjusted body weight in patients with a BMI ≥40 kg/m^2 (Ref).

Dosing: Older Adult

Refer to adult dosing; reduced doses may be required.

Dosing: Pediatric

(For additional information see "Etomidate: Pediatric drug information")

Anesthesia, induction

Anesthesia, induction:

Infants: Very limited data available: IV: 0.2 to 0.3 mg/kg/dose as a single dose; dosing based on a prospective, observational trial and two pharmacokinetic studies (Ref).

Children <10 years: Limited data available: IV: 0.2 to 0.3 mg/kg/dose as a single dose (Ref).

Children ≥10 years and Adolescents: Usual dose: IV: 0.3 mg/kg/dose as a single dose; range: 0.2 to 0.6 mg/kg/dose.

Procedural sedation

Procedural sedation: Limited data available; dosing variable: Infants ≥6 months, Children, and Adolescents: IV: Usual initial dose: 0.2 to 0.3 mg/kg/dose prior to procedure (reported range: 0.1 to 0.4 mg/kg/dose); repeat doses may be needed depending on patient response and duration of the procedure; repeat doses vary in the literature and range from 0.1 to 0.2 mg/kg/dose; reported maximum total dose ranged from 0.3 to 0.6 mg/kg per procedure (Ref). **Note**: Etomidate does **not** have analgesic properties; additional therapy (eg, fentanyl) may be needed for painful procedures (Ref).

Rapid sequence intubation

Rapid sequence intubation (RSI): Limited data available; Infants, Children, and Adolescents: IV, Intraosseous: 0.2 to 0.4 mg/kg/dose as a single dose; maximum dose: 20 mg/dose (Ref). **Note:** Not recommended for patients in septic shock due to an association between transient adrenocortical suppression and increased mortality rate (Ref).

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution, risk of toxicity is greater in patients with renal impairment.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Central nervous system: Myoclonus (33%)

Endocrine & metabolic: Adrenal suppression

Gastrointestinal: Nausea, vomiting (on emergence from anesthesia)

Local: Pain at injection site (30% to 80%)

Neuromuscular & skeletal: Musculoskeletal disease (transient skeletal movements)

Ophthalmic: Nystagmus

1% to 10%: Gastrointestinal: Hiccups

<1%, postmarketing, and/or case reports: Apnea, bradycardia, cardiac arrhythmia, decreased cortisol (decreased cortisol synthesis), hypertension, hyperventilation, hypotension, hypoventilation, laryngospasm, muscle spasm (masseter; Bozeman 2002), tachycardia

Contraindications

Hypersensitivity to etomidate or any component of the formulation

Warnings/Precautions

Concerns related to adverse effects:

• Adrenal steroid production: Etomidate inhibits 11-B-hydroxylase, an enzyme important in adrenal steroid production. A single induction dose blocks the normal stress-induced increase in adrenal cortisol production for 6 to 8 hours, up to 24 hours in elderly and debilitated patients. Continuous infusion of etomidate for sedation in the ICU may increase mortality because patients may not be able to respond to stress. Administration by continuous infusion is not recommended by the manufacturer. No increase in mortality has been identified with a single dose for induction of anesthesia (McPhee 2013).

Disease-related concerns:

• Heart failure: In a scientific statement from the American Heart Association, etomidate has been determined to be an agent that may exacerbate underlying myocardial dysfunction (magnitude: moderate) (AHA [Page 2016]).

• Renal impairment: Risk of toxicity is greater in patients with renal impairment; use with caution and monitor renal function.

Special populations:

• Older adult: May induce cardiac depression in elderly patients, especially those with hypertension; may require lower doses.

• Pediatric neurotoxicity: In pediatric and neonatal patients <3 years and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on child or fetal brain development and may contribute to various cognitive and behavioral problems. Epidemiological studies in humans have reported various cognitive and behavioral problems, including neurodevelopmental delay (and related diagnoses), learning disabilities, and ADHD. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. No specific anesthetic/sedative has been found to be safer. For elective procedures, risks versus benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed (FDA 2016).

Other warnings/precautions:

• Appropriate use: When considering use, weigh etomidate hemodynamic properties against the high frequency of transient skeletal muscle movements.

• Experienced personnel: According to the manufacturer, etomidate should only be administered by experienced personnel trained in the administration of general anesthetics and in the management of complications encountered during the conduct of general anesthesia. Consult local regulations and individual institutional policies and procedures.

Warnings: Additional Pediatric Considerations

Adrenal suppression has been documented with etomidate use, even after a single dose. Cortisol concentrations decrease quickly after the induction dose, lasting up to 8 hours in healthy adults and up to 24 hours in pediatric surgical patients; however, 2 prospective pediatric studies have reported no adverse clinical outcomes due to the adrenal suppression (Du 2015; Gu 2015). However, a retrospective study of pediatric patients admitted for meningococcal sepsis found patients receiving a single etomidate dose for rapid sequence intubation (n=23) had significantly decreased cortisol concentrations for at least 24 hours, and higher ACTH and 11-deoxycortisol concentrations compared to those who did not receive etomidate. In this study, 8 patients died due to hemodynamic failure, 7 of whom had received etomidate. While the role of etomidate on mortality is unclear in these cases, weigh the risk versus benefit prior to use and if use deemed necessary use with caution (den Brinker 2008). Guidelines suggest not to use etomidate in pediatric patients with sepsis or septic shock (SCCM [Weiss 2020]).

In pediatric and neonatal patients <3 years of age and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on the child's or fetus' brain development and may contribute to various cognitive and behavioral problems; the FDA is requiring warnings be included in the manufacturer's labeling for all general anesthetic/sedative drugs. Multiple animal species studies have shown adverse effects on brain maturation; in juvenile animals, drugs that potentiate GABA activity and/or block NMDA receptors for >3 hours demonstrated widespread neuronal and oligodendrocyte cell loss along with alteration in synaptic morphology and neurogenesis. Epidemiological studies in humans have reported various cognitive and behavioral problems including neurodevelopmental delay (and related diagnoses), learning disabilities, and attention-deficit/hyperactivity disorder. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. Further studies are needed to fully characterize findings and ensure that these findings are not related to underlying conditions or the procedure itself. No specific anesthetic/sedative has been found to be safer. For elective procedures, risk vs benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed (FDA 2016).

Some dosage forms may contain propylene glycol; in neonates, large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous:

Amidate: 2 mg/mL (10 mL, 20 mL) [contains propylene glycol]

Generic: 2 mg/mL (10 mL, 20 mL)

Solution, Intravenous [preservative free]:

Generic: 2 mg/mL (10 mL [DSC], 20 mL [DSC])

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Amidate Intravenous)

2 mg/mL (per mL): $0.68

Solution (Etomidate Intravenous)

2 mg/mL (per mL): $0.27 - $1.18

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Tomvi: 2 mg/mL (10 mL, 20 mL) [contains propylene glycol]

Generic: 2 mg/mL (10 mL, 20 mL)

Administration: Adult

IV: Administer IV push over 30 to 60 seconds. Solution is highly irritating; avoid administration into small vessels; in some cases, preadministration of lidocaine may be considered.

Administration: Pediatric

IV: Administer IV push over 30 to 60 seconds; very irritating; avoid administration into small vessels on the head or dorsum of the hand (Ref); preadministration of lidocaine may be beneficial (Ref).

Storage/Stability

Store at 20°C to 25°C (68°F to 77°F).

Use: Labeled Indications

General anesthesia: Induction of general anesthesia; as a supplement to subpotent anesthetic agents during maintenance of anesthesia for short operative procedures (eg, rapid sequence intubation, dilation and curettage, cervical conization).

Use: Off-Label: Adult

Cushing syndrome; Procedural sedation

Medication Safety Issues

Sound-alike/look-alike issues:

Etomidate may be confused with etidronate

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (anesthetic agent, general, inhaled and IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Disulfiram: May increase adverse/toxic effects of Products Containing Propylene Glycol. A disulfiram reaction may occur. Management: Consider avoiding concomitant use of disulfiram and products that contain propylene glycol. Note that not all preparations or dosage forms of a specific drug will contain propylene glycol. *Risk D: Consider Therapy Modification*

Secnidazole: Products Containing Propylene Glycol may increase adverse/toxic effects of Secnidazole. *Risk X: Avoid*

Pregnancy Considerations

Etomidate crosses the placenta (Esener 1992; Gregory 1991).

Based on animal data, repeated or prolonged use of general anesthetic and sedation medications that block N-methyl-D-aspartate receptors and/or potentiate GABA activity may affect brain development. Evaluate benefits and potential risks of fetal exposure to etomidate when duration of surgery is expected to be >3 hours (Olutoye 2018).

Use of etomidate for induction of anesthesia prior to cesarean delivery has been described (Downing 1979; Houlton 1978; Regaert 1984). However, other agents are more commonly used (ACOG 209 2019).

The ACOG recommends that pregnant women should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 775 2019).

Breastfeeding Considerations

Etomidate is present in breast milk (Esener 1992).

Following use of etomidate as an induction agent in 40 women undergoing cesarean delivery, etomidate was present in the colostrum at both 30 minutes and 2 hours after maternal injection. Etomidate was not present in breast milk 4 hours after the maternal dose (Esener 1992).

The manufacturer recommends that caution be exercised when administering etomidate to breastfeeding females; however, use is considered acceptable (Dalal 2014). The Academy of Breast Feeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Monitoring Parameters

Cardiac monitoring; blood pressure; renal function (in renal impairment); for treatment of Cushing syndrome, monitor serum cortisol levels every 4 to 6 hours (ES [Nieman 2015]).

Mechanism of Action

Ultra-short-acting nonbarbiturate general anesthetic (benzylimidazole) used for rapid induction of anesthesia with minimal cardiovascular effects; produces EEG burst suppression at high doses. Decreases endogenous cortisol synthesis via inhibition of 11-beta-hydroxylase (Pence 2022).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: 30 to 60 seconds

Peak effect: 1 minute

Duration: Dose dependent: 2 to 3 minutes (0.15 mg/kg dose); 3 to 5 minutes (0.3 mg/kg dose); rapid recovery is due to rapid redistribution

Distribution: V_d: 2 to 4.5 L/kg

Protein binding: 76%; decreased protein binding resulting in an increased percentage of “free” etomidate in patients with renal failure or hepatic cirrhosis

Metabolism: Hepatic and plasma esterases

Half-life elimination: Terminal: 2.6 to 3.5 hours

Time to peak, serum: 7 minutes

Excretion: Urine ~75% (80% as metabolite; 2% as unchanged drug)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Hepatic function impairment: V_d and elimination half-life increase 2-fold in patients with cirrhosis compared with healthy subjects.

Older adult: V_d, total clearance, and plasma protein binding are decreased in elderly patients.

Patient Counseling Points

(For additional information see ”Etomidate: Patient drug information”)

What is this drug used for?

• It is used for anesthesia before or during some procedures or surgery.

• It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Twitching

• Pain where the shot was given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight

• Trouble breathing, slow breathing, or shallow breathing

• Fast, slow, or abnormal heartbeat

• Muscle stiffness

• Trouble controlling body movements

• Not able to control eye movements

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Etomidate lipuro;
- (AR) Argentina: Midatus;
- (AT) Austria: Etomidat lipuro | Hypnomidate;
- (BE) Belgium: Etomidate Braun Pharma | Etomidate lipuro | Hypnomidate;
- (BG) Bulgaria: Hypnomidate;
- (BR) Brazil: Eptomydex | Etomidato | Hypnomidate;
- (CL) Chile: Endatal | Etomidato | Midatus;
- (CN) China: Etomidate lipuro | Fu er li;
- (CO) Colombia: Endatal | Etomidato Lipuro | Troymidate;
- (CZ) Czech Republic: Hypnomidate | Radenarcon;
- (DE) Germany: Etomidat lipuro | Hypnomidate | Radenarcon;
- (EC) Ecuador: Etomidato Lipuro;
- (EE) Estonia: Etomidat lipuro | Hypnomidate;
- (ES) Spain: Etomidato Lipuro | Hypnomidate | Sibul;
- (FR) France: Etomidate lipuro | Hypnomidate;
- (GB) United Kingdom: Etomidate lipuro | Hypnomidate;
- (GR) Greece: Hypnomidate;
- (HU) Hungary: Etomidat lipuro;
- (ID) Indonesia: Etomidate lipuro;

- (IN) India: Eldetec | Troymidate;
- (KR) Korea, Republic of: Etomidate lipuro;
- (KW) Kuwait: Etomidate lipuro;
- (LB) Lebanon: Hypnomidate;
- (LT) Lithuania: Etomidate lipuro | Hypnomidate;
- (LU) Luxembourg: Etomidat lipuro;
- (LV) Latvia: Etomidate lipuro | Hypnomidate;
- (MA) Morocco: Hypnomidate;
- (MX) Mexico: Endatal | Etomidato | Hypnomidate;
- (MY) Malaysia: Etomidate lipuro;
- (NL) Netherlands: Etomidaat-lipuro | Hypnomidate;
- (NO) Norway: Hypnomidate;
- (NZ) New Zealand: Etomidate lipuro | Hypnomidate;
- (PA) Panama: Hypnomidate;
- (PK) Pakistan: Etomidate lipuro;
- (PL) Poland: Etomidate lipuro | Hypnomidate | Radenarcon;
- (PR) Puerto Rico: Amidate;
- (PT) Portugal: Etomidato Lipuro | Hypnomidato;
- (QA) Qatar: Etomidate-Lipuro | Hypnomidate;
- (RO) Romania: Etomidat lipuro;
- (SG) Singapore: Etomidate lipuro;
- (SI) Slovenia: Etomidat lipuro | Hypnomidate;
- (SK) Slovakia: Hypnomidate;
- (TH) Thailand: Etomidate Braun Pharma | Etomidate lipuro | Hypnomidate;
- (TN) Tunisia: Hypnomidate;
- (TR) Turkey: Etomidate lipuro | Hypnomidate;
- (TW) Taiwan: Etomidate Braun Pharma | Hypnomidate;
- (UY) Uruguay: Etomidato | Etomidato cristalia | Hypnomidate;
- (ZA) South Africa: Hypnomidate;
- (ZW) Zimbabwe: Hypnomidate

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